brighter than the 2nd order.

17. Ans: C

Estimate the mean wavelength of visible light to be 500 nm.

Minimum angle of resolution of the telescope, $\theta = \dfrac{\lambda}{b} = \dfrac{500 \times 10^{-9}}{3.0} = 1.6667 \times 10^{-7}$ rad

(note that this angle is very small)

Let's apply the small angle approximation and "$S = r\theta$".

S is the minimum separation between the stars.

r = 20 light years = $20(3.0 \times 10^{8})(365 \times 24 \times 60 \times 60) = 1.892 \times 10^{17}$ m s^{-1}

$\theta = 1.6667 \times 10^{-7}$ rad

$\therefore$ $S = 3 \times 10^{10}$ m

18. Ans: C